Cow Health Monitoring — Data Schema & Dashboard Design

This document defines the **data schema** for storing ESP32 wearable sensor data and a **dashboard layout** (Grafana/React) for farmers and vets to visualize cow health trends.

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1. Data Schema

a) Core collections (MongoDB / TimescaleDB / InfluxDB)

`sensor_data`

- **device_id**: string (unique ID per cow collar)
- **timestamp**: datetime (UTC)
- **accX, accY, accZ**: float (m/s^2 aggregates)
- **temp**: float (°C, body/skin temperature)
- **battery**: float (voltage)
- **rumination_events**: int (number of chew events per sampling window)
- **status_flags**: array ["estrus_flag", "fever_flag", ...] (optional)

`alerts`

- **alert_id**: string
- **device_id**: string
- **timestamp**: datetime
- **type**: string ("LOW_BATTERY", "HEAT_DETECTED", "FEVER")
- **severity**: string ("info", "warning", "critical")
- **message**: string

`devices`

- **device_id**: string
- **cow_id**: string
- **farm_id**: string
- **last_seen**: datetime
- **firmware_version**: string

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2. Dashboard Layout (Grafana / Custom React App)

a) Farm Overview Page

- **Map / Barn layout view** with colored status indicators per cow.

- **Summary cards**:
- Total cows monitored
- Cows in heat today
- Suspected sick cows
- Battery warnings

b) Cow Detail Page

- **Header:** Cow ID, age, lactation stage, device status.
- **Charts:**
- **Temperature trend** (line chart, 24h & 7d views).
- **Rumination events** (bar chart per hour/day).
- **Activity (acceleration magnitude)** (line chart).
- **Battery voltage** (line chart).
- **Alerts timeline** (heat, fever, low battery).
- **Vet notes / interventions log**.

c) Heat Detection Dashboard

- Table view of cows likely in estrus with probability/confidence.
- Activity spike chart (relative to baseline).
- Alert export (WhatsApp/SMS/email integration).

d) Health Alerts Dashboard

- List of cows with fever or abnormal rumination drop.
- Severity-based color coding.
- Drill-down into affected cows.

e) Device Maintenance Dashboard

- Firmware version distribution.
- Battery health across devices.
- Offline devices list.

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3. Alerts & Notifications

- **Trigger rules:**
- Estrus: activity > baseline + threshold & rumination drop.
- Fever: temp > baseline + 1.5 °C for >6h.
- Low battery: < 3.5 V.
- **Delivery channels:** WhatsApp, SMS, Email, In-app push.
- **Action buttons:** "Mark checked", "Schedule vet visit".

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4. Tech stack suggestions

- **Backend:** Node.js + MQTT broker + MongoDB/InfluxDB.
- **Frontend:** React + Grafana for charts.
- **Deployment:** Docker + farm edge server with optional cloud sync.

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5. Example JSON Payload from ESP32


```json
{
  "device_id": "cow01",
  "timestamp": "2025-09-23T12:00:00Z",
  "accX": -0.15,
  "accY": 0.98,
  "accZ": 9.81,
  "temp": 38.4,
  "battery": 3.9,
  "rumination_events": 12,
  "status_flags": ["estrus_candidate"]
}
```


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6. Roadmap

1. Phase 1: Farm overview + cow detail dashboard.
2. Phase 2: Heat detection dashboard with alerts.
3. Phase 3: Health alert predictions + vet intervention logs.
4. Phase 4: Integration with institutional buyers (milk yield, quality).
